

Failure-Complete Evaluation of Semantic Rescue in Few-Shot Tabular Self-Training

Anonymous Authors

Abstract—Semantic representations can provide side information for few-shot tabular learning, but an adaptive rescue route is useful only when it improves the downstream predictor under a fair information boundary. We test whether strictly out-of-fold semantic evidence can rescue unlabeled rows rejected by a matched tree self-training system without using candidate or evaluation labels. ProtoOOF-Tree-v1 forms leave-one-pair-out class prototypes from a fixed T0-3B representation, makes every activation decision from labeled support alone, and otherwise returns the tree endpoint exactly. The frozen kill-screen covers two OpenML datasets, three shot counts, and three seeds: 18 episodes and 72 ordered method rows spanning the tree anchor, semantic route, no-semantics control, and row-permuted control. The semantic route matches the tree in all 18 episodes, so the overall paired ROC-AUC difference is +0.0000000. Mean separation from the no-semantics and permuted controls is +0.0009668 and -0.0005258, respectively, both below the registered +0.0100 threshold. The main route records 0 gate acceptances and 0 rescued rows; the two controls record 1/50 and 2/100. Five of ten promotion conditions fail, and the frozen decision is NO_GO. The result bounds this particular leakage-controlled gate and transfer path; it does not establish that semantic tabular learning is generally ineffective.

Keywords—*adaptive decision-making, few-shot tabular learning, self-training, semantic representations, reliable artificial intelligence, negative results, data mining*

I. INTRODUCTION

Few-shot tabular classification combines scarce labels with heterogeneous features. Tree ensembles remain difficult to displace under controlled tabular comparisons [1], [2], while language-model approaches use feature names, values, or row serializations as prior information [3]–[6]. The engineering question is therefore additive rather than cosmetic: does a semantic path improve the final tree, or does it merely produce plausible scores and an active-looking module?

Self-training can enlarge a small labeled support set, but feedback can also reinforce early mistakes [7]. Confidence selection couples pseudo-label quality with coverage [8]. A semantic rescue can appear successful because it contributes useful information, repeats the tree ordering in another space, or follows an unfair information path. We therefore require support-only gating, matched controls, an immutable tree anchor, and retention of every registered episode.

We ask three questions. RQ1 asks whether the semantic route improves mean ROC-AUC over the matched tree. RQ2 asks whether it separates from feature-only and row-permuted controls. RQ3 asks whether it passes a registered reliability gate. The decision is conjunctive: an attractive intermediate score, one favorable episode, or exact no-harm fallback cannot substitute for downstream improvement.

We evaluate ProtoOOF-Tree-v1, a deliberately bounded route from a fixed semantic representation to a tree self-training endpoint. The method constructs semantic class prototypes in leave-one-pair-out folds, compares semantic and tree-native support evidence, and can append only candidates that the tree route rejected. Candidate and evaluation labels are absent from representation construction, gating, thresholding, and rescue. When support evidence is insufficient, the complete training arrays and final predictions must equal the tree anchor exactly.

The contribution is an evaluation contract rather than a claim that a single component is novel in isolation. The contract links an incremental endpoint estimand, out-of-fold evidence, two semantic-dependence controls, exact fallback, and a failure-complete stop rule. Its practical value is to make

a negative development decision reproducible before a larger confirmation consumes more data, computation, or interpretation.

The answer in the frozen study is negative. ProtoOOF-Tree-v1 never activates rescue and exactly matches TreeSelfTrain-S in all 18 episodes. The result is not presented as statistical equivalence or as a general rejection of semantic representations. It is evidence that this particular representation, gate, and transfer rule should not be promoted under the registered criteria.

II. RELATED WORK

Strong tabular baselines and priors. XGBoost established a scalable regularized boosting framework [1]. Tabular foundation models take a different route: TabPFN and TabICL learn priors for in-context prediction [9], [10]. These lines of work motivate both a competitive tree anchor and disciplined accounting of what information an auxiliary route receives.

Language-derived tabular information. TabLLM serializes rows for language-model prediction [3]; TransTab learns representations transferable across tables [4]; and FeatLLM converts language-model rules into features for a simpler learner [5]. T0 supplies the fixed multitask-prompted representation used here [6]. Our claim is narrower: any auxiliary semantics must be evaluated at the final deployed endpoint with matched information access.

Selection and negative tests. Classic pseudo-labeling uses model predictions as training targets [7], whereas later work treats confidence thresholds as explicit coverage-error decisions [8]. VIME shows how semi-supervised assumptions must suit heterogeneous tabular data [11]. These results motivate support-only selection, but they do not remove the need to test whether semantic content is responsible for an observed change.

Broader benchmark evidence reinforces this caution. Tree-based models remain strong on many typical tables [12], and LightGBM and CatBoost embody distinct inductive and systems choices [13], [14]. FixMatch illustrates how confidence controls data inclusion in semi-supervised learning [15]. The common lesson is that a selection mechanism must

be judged through its final learner, not through confidence or activity alone.

III. METHOD AND FAIR INFORMATION PROTOCOL

Episode and endpoint. Each registered episode is a (dataset, shot, seed) triple. The binary datasets are OpenML credit-g (ID 31) and spambase (ID 44); shots are 5, 10, and 20, and seeds are 0, 9, and 19. TreeSelfTrain-S is the fixed anchor. All four routes share labeled support, candidate pool, evaluation set, preprocessing, split, and ROC-AUC endpoint.

TreeSelfTrain-S is the deployment anchor. It consumes the registered support, selects its own pseudo-labeled candidate rows, assigns targets and reliability weights, and fits the final tree. ProtoOOF-Tree-v1 is not allowed to replace the anchor's accepted pseudo-labeled rows or reorder the anchor training prefix. It may append only a rescue set $A(d, K, s)$ drawn from rows that the tree rejected. This constraint turns the comparison into an incremental-information test: any endpoint difference must enter through the appended semantic rescue.

Feature-only semantic representation. Every row is serialized without a label token, label name, metric, or downstream result and embedded with one sealed T0-3B revision. The semantic cache is fixed before the report is opened. Main and control routes share the episode split, support pairs, cache identity, and numerical pipeline; the study does not adapt prompts after seeing outcomes.

Out-of-fold prototype evidence. Support examples are paired across the two classes. For each held-out pair, the remaining support representations define two class prototypes. Relative similarity supplies a signed score for the held-out row. Repeating this operation gives one semantic score per support row without allowing that row to define the prototypes used to score it.

The activation gate compares semantic and tree-native evidence using only the held-out support scores. The candidate confidence threshold is the median absolute support prototype score. There is no threshold search, dataset-specific adjustment, or performance-driven sign flip. If the gate accepts, a rejected tree candidate is eligible only when the semantic and tree predictions disagree in the prespecified direction and the semantic magnitude meets the support-derived threshold. An eligible row receives the registered semantic target and a reliability weight capped by the frozen alpha rule.

Exact fallback. If the gate rejects, the eligible rescue set is empty, or a registered numerical failure occurs, the route cannot be relabeled as a success. With no accepted rescue, every training row, target, reliability weight, ordering decision, and final prediction must match TreeSelfTrain-S exactly. Alpha zero is checked separately. This is an implementation-safety invariant, not evidence of accuracy.

ProtoOOF-NoSemantics replaces the semantic representation with a feature-only nonsemantic representation while preserving folds, gate, threshold logic, and downstream fitting. ProtoOOF-Permuted applies a fixed row permutation while preserving the representation shape and computation. The ordered four-method inventory is therefore TreeSelfTrain-S, ProtoOOF-Tree-v1, ProtoOOF-NoSemantics, and

ProtoOOF-Permuted. The controls do not prove a complete causal decomposition, but they provide necessary tests: an effect that does not separate from either control cannot be attributed to the intended semantic alignment.

The primary estimand is the paired episode difference $AUC(Main) - AUC(Tree)$. Because methods share the same split and seed, this quantity isolates the incremental endpoint effect of the complete rescue route. It is averaged across all 18 episodes. No success-only denominator, best seed, or complete-case filter is permitted.

TABLE I. MATCHED INFORMATION ACCESS

Route	Semantic content	Candidate labels	Endpoint
Tree	No	No	Tree
Main	Fixed T0-3B	No	Tree after accepted rescue or exact fallback
NoSem	Removed	No	Matched control tree
Perm	Row-permuted	No	Matched control tree

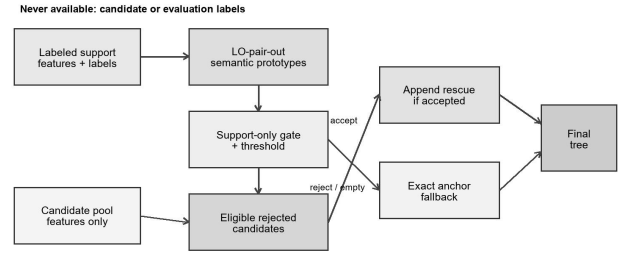


Figure 1. Registered adaptive rescue flow. Candidate features may enter the fixed semantic cache, but candidate and evaluation labels never enter prototypes, gating, thresholding, or rescue. A rejected or empty rescue returns the anchor exactly.

IV. FROZEN EXPERIMENTAL PROTOCOL

The formal development scope contains two binary OpenML datasets: credit-g (OpenML 31) and spambase (OpenML 44). The source audit records 1,000 rows and 20 features for credit-g, including 13 categorical and seven continuous features, with class counts 700/300. Spambase contains 4,601 rows, 57 continuous features, and class counts 2,788/1,813. These differences make the pair a small but useful contrast between mixed-type financial data and dense continuous frequency features. They do not constitute broad task coverage.

Registered shot counts are 5, 10, and 20; registered seeds are 0, 9, and 19. Their Cartesian product gives 18 episodes. Each episode emits one ordered record for each of four methods, for 72 method rows. Dataset bytes, normalized manifests, label audits, split identities, study source, execution protocol, and report builder are content-addressed. The released report records zero execution failures. Table I records the complete study grid and evidence boundary.

Fairness is defined by common episode identity and a bounded method-only difference. All routes share support and candidate pools, final tree family, metric, row-order contract, and failure policy. ProtoOOF variants also share fold construction, support-only gate, threshold rule, and numerical pipeline. Only the representation or control transformation changes.

The development screen returns only GO_TO_INDEPENDENT_CONFIRMATION or NO_GO. GO requires every registered conjunct: no failed method row; an overall main-minus-tree mean of at least 0.02 ROC-AUC; nonnegative mean and median on both datasets; at least 0.01 mean separation from each control; a positive main-minus-tree mean for at least two shot levels; exact fallback and alpha-zero equivalence; nonempty rescue evidence on both datasets; and zero pseudo-label collapse. The use of a conjunction matters. Passing no-harm while failing incremental benefit is a safe implementation result, not a successful learning result. Table III pairs every condition with its frozen outcome.

Held-out ROC-AUC is the episode endpoint. It evaluates ranking without selecting a test threshold and is well defined for the registered binary tasks, but it does not measure probability calibration or a deployment cost function. The same evaluation rows and metric implementation are used for every method within an episode. Paired differences are formed before aggregation, preserving the dataset/shot/seed match. Means by dataset and shot are secondary components of the registered gate rather than separately optimized endpoints.

A method-row failure is not imputed with a favorable value and is not dropped. Instead, the row remains in the 72-record inventory with its error code, and the no-failure conjunct fails. Numerical failure, gate rejection, and empty rescue are deliberately different states: failure invalidates the episode for promotion, whereas rejection or empty rescue invokes an exact, testable anchor fallback. The formal run contains the latter state for all 18 main-route episodes and no execution failure.

Leakage prevention is checked at several interfaces. The representation cache persists feature-only text and embeddings, with no label token, label name, evaluation metric, or downstream result. A held-out support pair does not contribute to the prototypes used to score that pair. Candidate features are unavailable to the support activation decision, and candidate labels are unavailable throughout. Evaluation labels are opened only by the common endpoint evaluator. These controls do not rule out prior knowledge of a public dataset inside a pretrained model; they rule out direct use of the registered episode labels by the transfer pipeline.

Fallback equivalence is stronger than comparing two rounded metrics. The receipts require the training rows, targets, reliability weights, ordering, and predictions to be array-exact to TreeSelfTrain-S when rescue is absent. A separate alpha-zero route verifies that the semantic append path reduces to the same anchor when its weight is disabled. The formal report records both equivalence requirements as satisfied in every applicable row. This allows the observed main-tree zero to be traced to a specific invariant rather than to rounding or insufficient metric precision.

The hashes recorded in the evidence manifest identify the exact report, method ledger, episode-difference ledger, and summary used for this paper. Hashing does not prove that every scientific design choice is correct, but it prevents a different result file from being substituted after interpretation. The complete 18-episode ledger is therefore reproduced in Table II rather than summarized by a favorable subset.

TABLE II. COMPLETE FROZEN EPISODE LEDGER (ROC-AUC AND GATE/RESCUE COUNTS)

Dataset	Shot	Seed	Tree	Main	NoSem	Perm	M G/R	N G/R	P G/R
credit-g	5	0	0.61500	0.61500	0.61500	0.61292	0/0	0/0	1/50
credit-g	5	9	0.49071	0.49071	0.49071	0.50226	0/0	0/0	1/50
credit-g	5	19	0.59220	0.59220	0.59220	0.59220	0/0	0/0	0/0
credit-g	10	0	0.57256	0.57256	0.57256	0.57256	0/0	0/0	0/0
credit-g	10	9	0.63952	0.63952	0.63952	0.63952	0/0	0/0	0/0
credit-g	10	19	0.57244	0.57244	0.57244	0.57244	0/0	0/0	0/0
credit-g	20	0	0.69417	0.69417	0.69417	0.69417	0/0	0/0	0/0
credit-g	20	9	0.69077	0.69077	0.69077	0.69077	0/0	0/0	0/0
credit-g	20	19	0.67524	0.67524	0.67524	0.67524	0/0	0/0	0/0
spambase	5	0	0.83315	0.83315	0.83315	0.83315	0/0	0/0	0/0
spambase	5	9	0.82425	0.82425	0.80685	0.82425	0/0	1/50	0/0
spambase	5	19	0.83195	0.83195	0.83195	0.83195	0/0	0/0	0/0
spambase	10	0	0.88925	0.88925	0.88925	0.88925	0/0	0/0	0/0
spambase	10	9	0.88486	0.88486	0.88486	0.88486	0/0	0/0	0/0
spambase	10	19	0.82974	0.82974	0.82974	0.82974	0/0	0/0	0/0
spambase	20	0	0.91011	0.91011	0.91011	0.91011	0/0	0/0	0/0
spambase	20	9	0.91171	0.91171	0.91171	0.91171	0/0	0/0	0/0
spambase	20	19	0.90035	0.90035	0.90035	0.90035	0/0	0/0	0/0

Tree = TreeSelfTrain-S; Main = ProtoOOF-Tree-v1; NoSem = ProtoOOF-NoSemantics; Perm = ProtoOOF-Permuted; G/R = gate accepted/rescued rows. All 72 method rows succeeded; all collapse indicators are zero.

V. FROZEN RESULTS

RQ1 - endpoint. ProtoOOF-Tree-v1 matches TreeSelfTrain-S in every registered episode. The overall main-minus-tree mean is +0.0000000; credit-g and spambase each have mean and median 0.0000; and the number of shot levels with a positive mean is 0. Main records 0 gate acceptances and 0

rescued rows, so exact fallback explains the deterministic endpoint equality.

RQ2 - controls. Mean Main-NoSemantics is +0.0009668, whereas mean Main-Permuted is -0.0005258. Neither reaches the registered +0.0100 separation, and the latter has the wrong sign. NoSemantics records 1 accepted gate and 50 rescued rows; Permuted records 2 and 100. Activity in these controls does not establish semantic benefit.

Only three control contrasts depart from zero. On credit-g at five shots, the main-minus-Permuted differences are +0.0020833 for seed 0 and -0.0115476 for seed 9. On spambase at five shots and seed 9, the main-minus-NoSemantics difference is +0.0174028. Every other main-control contrast is zero. Because one permuted control is better than the main route and another is worse, and because all departures occur at five shots, the pattern is consistent with an unstable low-support gate. It does not identify instability as the unique cause; three departures are too few for a reliable distributional conclusion.

RQ3 - decision. Five of ten conjuncts fail: overall_main_delta_min, main_minus_no_semantics_min, main_minus_permuted_min, positive_shot_means_min_count, and rescue_required_both_datasets. All 72 method rows execute successfully, no pseudo-label collapse occurs, and exact fallback passes. Because all conditions are required, the frozen verdict remains NO_GO.

The exact main-tree zeros require a different interpretation from small paired estimates near zero. Here the main route does not add a candidate and the fallback contract forces identical arrays and predictions. The endpoint equality is therefore explained by the observed execution path. It should not be converted into a confidence interval for equivalence, and it provides no estimate of what would happen if the main gate activated.

The result is stronger than a claim that the mean is merely close to zero and weaker than a claim that all semantic rescue effects are zero. It is stronger because all 18 paired outcomes are generated by a verified exact fallback, not by cancellation of positive and negative estimates. It is weaker because the registered main gate never exposes the final tree to a main-route rescue. The experiment therefore falsifies the complete proposal as a usable promotion candidate: a mechanism that cannot activate under its own safety rule delivers no incremental benefit. It does not estimate the effect of forcibly bypassing that rule.

TABLE III. REGISTERED CONJUNCTIVE DECISION

Condition	Threshold	Observed	Result
Overall Main-Tree mean	$\geq +0.0200$	+0.0000	FAIL
Main-NoSem mean	$\geq +0.0100$	+0.0010	FAIL
Main-Perm mean	$\geq +0.0100$	-0.0005	FAIL
Dataset mean/median	both nonnegative	both 0.0000	PASS
Positive shot means	≥ 2 of 3	0	FAIL
Rescue coverage	both datasets	neither (main)	FAIL
Pseudo-label collapse	zero	zero	PASS
Failed method rows	zero	0	PASS
Exact registered rows	18/72	18/72	PASS
Exact fallback	all fallback exact	pass	PASS

VI. FAILURE ANALYSIS AND DECISION VALUE

The directly observed boundary is narrow. The main semantic comparison never satisfies its activation rule; exact fallback leaves the tree unchanged; and matched controls activate in three low-shot episodes without establishing the intended semantic alignment. These observations locate the failure within the tested path from representation through support evidence to candidate rescue.

Several explanations remain plausible but unverified. Leave-one-pair-out support evidence may be too variable at 5, 10, or 20 shots. The support-derived threshold may be conservative when semantic margins are compressed. The fixed T0-3B geometry may encode broad feature meaning without aligning with the tree error set. Because Main never activates, this study cannot distinguish these mechanisms.

The control pattern adds one sharper warning. A support-only gate can pass because of small-sample ordering noise even when semantic alignment has been removed or permuted. A future design should therefore estimate gate stability itself, for example through repeated support resampling, and should require the semantic route to beat control activation rates before candidate rescue is considered. This proposal is a design implication, not a result already tested here.

Negative evidence is useful when it changes a decision. Counting only accepted gates could have turned the three control activations into an apparently active method. Reporting only no-harm could have described 18 exact fallbacks as a successful semantic augmentation. Neither statement answers whether semantics improved the endpoint. The registered rule prevents both interpretations.

For practitioners, the present recommendation is simple: retain TreeSelfTrain-S under this protocol. The semantic route adds representation and gating machinery but produces the same final predictions. This is not a permanent rejection of language-derived features. It is a refusal to pay for an unvalidated pathway whose intended semantic control separation also fails.

The third lesson concerns controls. A no-semantics route checks whether generic feature geometry or calibration is sufficient for activation; a permuted route checks whether row alignment matters. Both should traverse the same code path and should be judged by prespecified contrasts. If a control activates, the correct response is not to delete that episode but to weaken the semantic attribution. Finally, failure handling belongs in the estimand. A reliable method is a system that returns a scientifically interpretable record for every registered unit, including rejection and failure states.

VII. LIMITATIONS AND THREATS TO VALIDITY

External validity is narrow. The screen covers two binary OpenML datasets, three shot counts, and three seeds. It can reject the registered mechanism but cannot characterize performance on high-cardinality categorical data, multiclass tasks, regression, domain-specific tables, or larger support sets. Three seeds per condition are insufficient for a precise sampling distribution; the decision is therefore threshold-based and failure-complete, not a claim of statistical equivalence.

Construct validity is limited by one fixed T0-3B revision, one feature-only serialization, one prototype score, one support gate, and one tree family. The no-semantics and permuted controls test necessary dependence on the intended representation, but they do not identify all sources of calibration, row-order sensitivity, or prototype instability. Because the registered development route never rescues, the

study cannot estimate the conditional quality of main-route rescued labels.

The protocol is transductive: unlabeled candidate features are available when the semantic cache and candidate scores are constructed. Deployments that require strictly inductive prediction have a different information boundary. The evidence package content-addresses the artifacts needed for the frozen result, but the manuscript does not infer unrecorded hardware, latency, energy, or full environment details. No efficiency claim is made.

This is a development kill-screen, not an independent confirmation. Changing thresholds, datasets, seeds, failure treatment, serialization, model revision, or estimand creates a new study. The current zero effects arise from exact fallback, not from a sampled distribution of small nonzero effects; no p-value, significance test, or equivalence margin is added after seeing the outcomes.

Reporting validity also has limits. The artifact chain preserves the frozen result and exposes the stop rule, but an external reader does not receive a new independent rerun through this paper. Dataset names and OpenML identifiers are disclosed for reproducibility, while anonymous review requires that personal repository paths and local machine identities remain absent. A post-review release should provide the nonidentifying manifests, runner, and report builder needed to reproduce the tables without exposing private infrastructure.

VIII. REPRODUCIBILITY AND FALSIFICATION

The evidence contract fixes two OpenML identifiers, three shot counts, three seeds, four ordered routes, and all failure semantics before interpretation. A reproduction begins from the registered split and fixed representation cache, runs TreeSelfTrain-S, Main, NoSemantics, and Permuted for each episode, and records ROC-AUC together with gate and rescue activity.

The evidential unit is the registered episode, not a selected successful run. A sufficient release links the frozen decision conditions, the 72-row ledger with explicit failure fields, episode contrasts, aggregate report, and a content-addressed manifest. A reviewer can then recompute every mean, median, activity count, control difference, and conjunctive decision without inferring missing rows from prose. Eighteen episode identities times four ordered methods must equal 72 rows, each registered unit must occur once, and every summary must be derivable from that ledger.

The checklist also separates cache completion, gate eligibility, rescue activity, held-out endpoint change, and the derived promotion verdict. The first three cannot license a predictive claim. Reporting all five levels prevents exact fallback from being mistaken for useful adaptation and makes the negative boundary falsifiable.

A new study may challenge this boundary, but it should be frozen independently. It must specify a replacement representation or gate, distribution-shift model, datasets, shots, seeds, controls, thresholds, and failure treatment before opening outcomes. A new GO requires stable rescue coverage,

dependence on semantic content under matched controls, and a positive final-tree effect across datasets and shot levels.

TABLE IV. FALSIFIABLE FOLLOW-UP HYPOTHESES

Hypothesis	Evidence needed to challenge the boundary
Support gate is underpowered	Pre-registered stronger support evidence activates stably on both datasets without label access
Semantic geometry misses tree errors	A frozen alternative representation separates from both controls and improves the tree
Transfer is the bottleneck	Accepted rescues are accurate and yield positive paired endpoint effects under the same anchor
Effect is distribution-specific	A preselected broader grid passes dataset and shot consistency conditions

IX. CONCLUSION

A failure-complete evaluation prevents mechanism activity from being confused with predictive value. ProtoOOF-Tree-v1 is leakage-controlled and falls back exactly, but it supplies no final-tree gain and no registered control separation in the frozen study. The decision is therefore NO_GO. Exact fallback is a safety property; only a reproducible endpoint improvement under matched controls can justify deployment. The result retains a narrow interpretation: it rejects this gate-and-transfer route while leaving other semantic representations and adaptive designs open to independently frozen tests.

AI Tool Use Disclosure - External language editing and OpenAI Codex (GPT-5 family, accessed September 2026) assisted with English phrasing, formatting, and consistency checks. These tools did not generate or alter experimental data. The author retained and verified the research design, code, frozen results, references, and interpretations.

REFERENCES

- [1] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, 2016, pp. 785-794. <https://doi.org/10.1145/2939672.2939785>
- [2] Y. Gorishniy, I. Rubachev, V. Khrulkov, and A. Babenko, "Revisiting deep learning models for tabular data," in Advances in Neural Information Processing Systems, vol. 34, 2021. https://proceedings.neurips.cc/paper_files/paper/2021/hash/9d86d83f925f2149e9edb0ac3b49229c-Abstract.html
- [3] S. Hegselmann, A. Buendia, H. Lang, M. Agrawal, X. Jiang, and D. Sontag, "TabLLM: Few-shot classification of tabular data with large language models," in Proc. AISTATS, PMLR 206, 2023, pp. 5549-5581. <https://proceedings.mlr.press/v206/hegselmann23a.html>
- [4] Z. Wang and J. Sun, "TransTab: Learning transferable tabular transformers across tables," in Advances in Neural Information Processing Systems, vol. 35, 2022. <https://doi.org/10.52202/068431-0210>
- [5] S. Han, J. Yoon, S. O. Arik, and T. Pfister, "Large language models can automatically engineer features for few-shot tabular learning," in Proc. ICML, PMLR 235, 2024, pp. 17454-17479. <https://proceedings.mlr.press/v235/han24f.html>
- [6] V. Sanh et al., "Multitask prompted training enables zero-shot task generalization," in Proc. ICLR, 2022. <https://openreview.net/forum?id=9Vrb9D0W14>
- [7] D.-H. Lee, "Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks," ICML Workshop on Challenges in Representation Learning, 2013. <https://openreview.net/pdf?id=3iGjhNmoG>
- [8] H. Vishwakarma, Y. Chen, S. S. S. Namburi Gnvv, S. J. Tay, R. K. Vinayak, and F. Sala, "Rethinking confidence scores and thresholds in pseudolabeling-based SSL," in Proc. ICML, PMLR 267, 2025, pp. 61582-61600. <https://proceedings.mlr.press/v267/vishwakarma25a.html>
- [9] N. Hollmann, S. Muller, K. Eggenberger, and F. Hutter, "TabPFN: A transformer that solves small tabular classification problems in a second," in Proc. ICLR, 2023. <https://openreview.net/forum?id=cp5Pvc16w8>
- [10] J. Qu, D. Holzmüller, G. Varoquaux, and M. Le Morvan, "Tab4CL: A tabular foundation model for in-context learning on large data," in Proc. ICML, PMLR 267, 2025, pp. 50817-50847. <https://proceedings.mlr.press/v267/qu25d.html>
- [11] J. Yoon, Y. Zhang, J. Jordon, and M. van der Schaar, "VIME: Extending the success of self- and semi-supervised learning to tabular domain," in Advances in Neural Information Processing Systems, vol. 33, 2020. <https://proceedings.neurips.cc/paper/2020/hash/7d97667a3e056acab9aaf653807b4a03-Abstract.html>
- [12] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on typical tabular data?" in Advances in Neural Information Processing Systems, vol. 35, 2022. <https://doi.org/10.52202/068431-0037>
- [13] G. Ke et al., "LightGBM: A highly efficient gradient boosting decision tree," in Advances in Neural Information Processing Systems, vol. 30, 2017. <https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bd9eb6b76fa-Abstract.html>
- [14] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: Unbiased boosting with categorical features," in Advances in Neural Information Processing Systems, vol. 31, 2018. <https://proceedings.neurips.cc/paper/2018/hash/14491b756b3a51daac41c24863285549-Abstract.html>
- [15] K. Sohn, D. Berthelot, N. Carlini, Z. Zhang, H. Zhang, C. A. Raffel, E. D. Cubuk, A. Kurakin, and C.-L. Li, "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in Advances in Neural Information Processing Systems, vol. 33, 2020. <https://proceedings.neurips.cc/paper/2020/hash/06964dce9addb1c5cb5d6e3d9838f733-Abstract.html>